

Impact of Anthropogenic and Agricultural Land Use on Nutrient Levels of the Dorset Lakes in the Muskoka River Watershed

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BACKGROUND

Across Ontario, climate change and anthropogenic development leading to land-use/cover change are exacerbating threats to water quality including eutrophication.¹ Eutrophication, driven by loading of limiting nutrients, mainly phosphorus (P) and nitrogen (N), can be detrimental to aquatic ecosystems as it directly influences the growth of algal blooms including toxic cyanobacteria, lake oxygen depletion and more.² Agricultural and urban (residential, commercial, industrial) land uses are leading contributors to nutrient loading.^{3,4} Lakes in the Muskoka River Watershed (MRW) are increasingly affected by algal blooms, endangering public health and lake ecosystems.⁵ Since the mid-1970s, the Dorset Lakes Monitoring Program (DLMP), part of the District of Muskoka Integrated Water Management Plan, has intensively studied water quality in eight MRW lakes to assess the impacts of these stressors.⁶ As land use changes continue to accelerate, monitoring the Dorset Lakes remains essential, as they serve as indicators of water quality trends in similar small lakes across the MRW.⁶

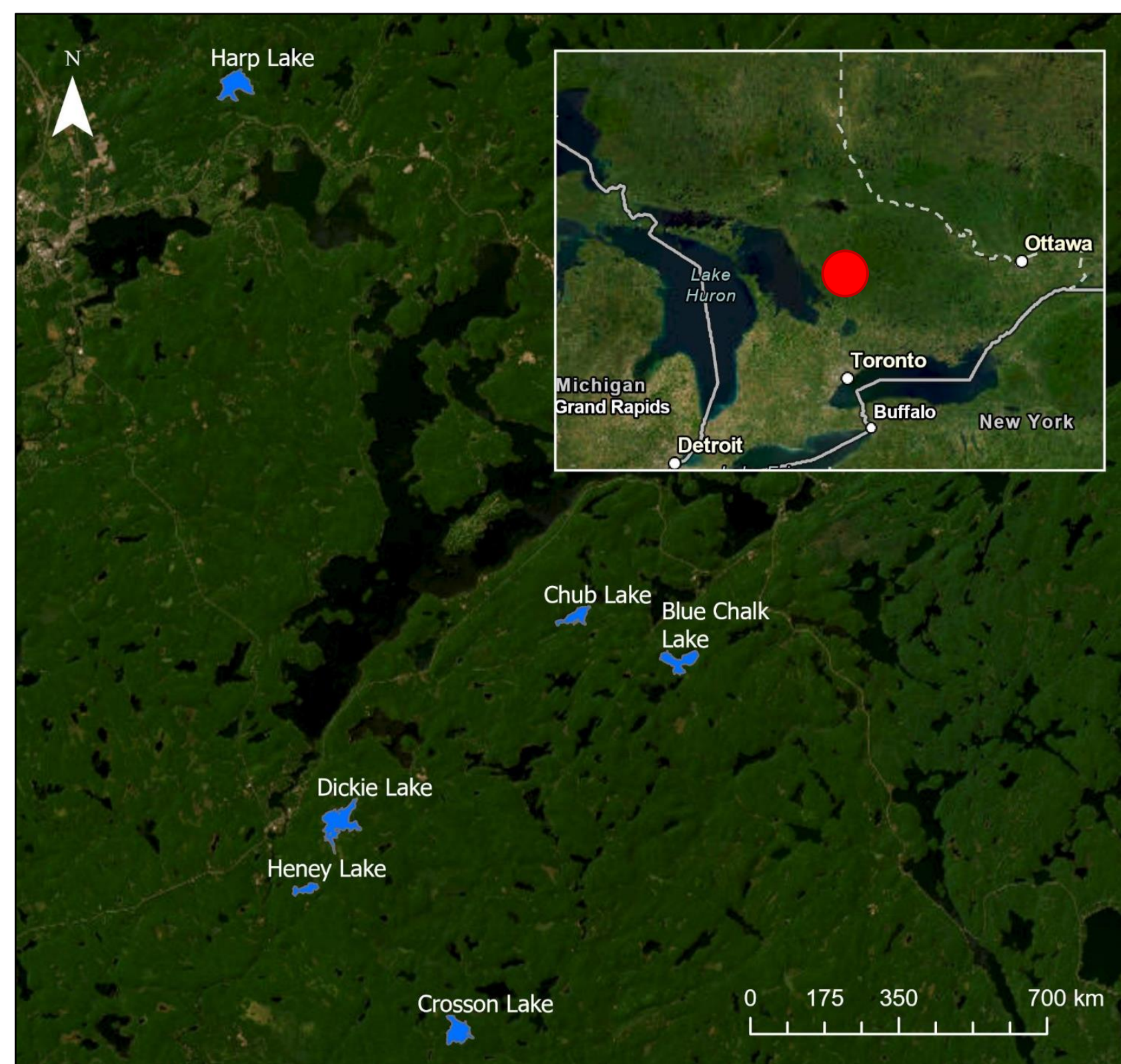
OBJECTIVES

This primary goal of this study is to evaluate whether agricultural and anthropogenic (urban/built-up) land use is influencing the concentrations of limiting nutrients in six of the lakes monitored by the DLMP.

Specifically, this study aims to:

1. Identify lakes that have experienced increases in total phosphorus (TP) and nitrogen (TN) concentrations over time, and those with the highest recent levels.
2. Calculate percent area of agricultural and anthropogenic land use within multiple buffer zones surrounding each lake.
3. Determine if there is a correlation between land use types and TP and TN levels in the lakes.

STUDY AREA



- Blue Chalk Lake: 50 ha
- Chub Lake: 33 ha
- Crosson Lake: 57 ha
- Dickie Lake: 224 ha
- Harp Lake: 72 ha
- Heney Lake: 21 ha

Figure 1. Location of the six DLMP lakes of interest in the MRW within the Dorset township, Muskoka District, Ontario, Canada.⁷

METHODS

Calculating Percent Area of Land Use Types - ArcGIS Pro

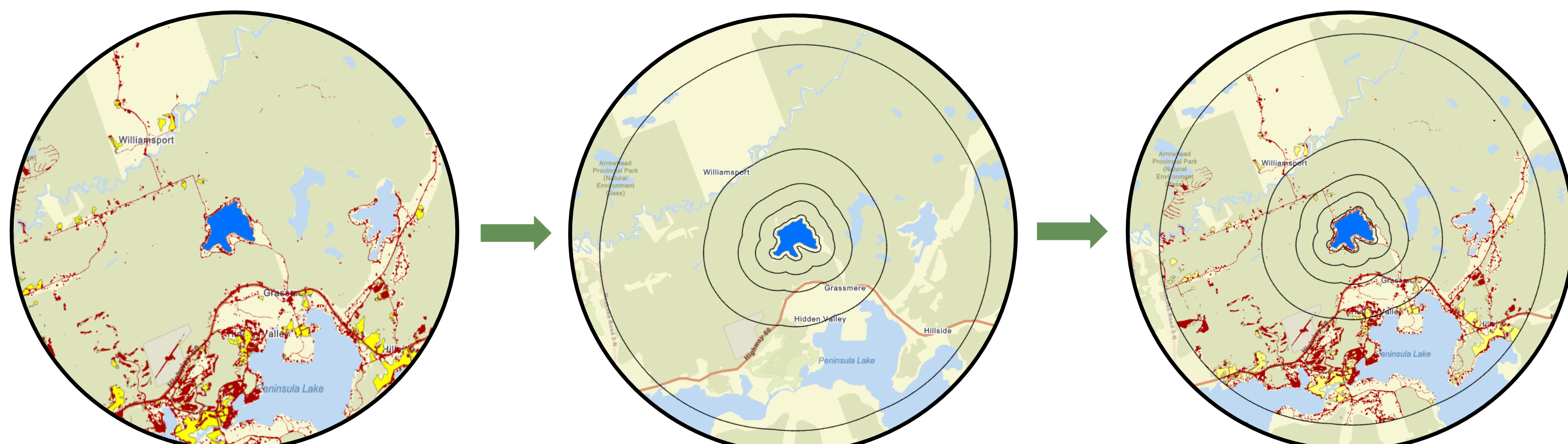


Figure 2. Workflow of land use area calculations in ArcGIS Pro, using Harp Lake as an example.

- Geospatial land use data (lakes, agricultural and anthropogenic) from the Muskoka GeoHub^{8,9} was used to first create buffer zones at 100m, 500m, 1km, 2km, and 5km around the lakes.
- The *Intersect* tool was applied to extract land cover data within each buffer zone.
- The *Summarize Statistics* tool was used to calculate the area of both land use types within the defined buffer distances.
- Using the area of anthropogenic and agricultural land use and the total area within each buffer zone, % area was calculated per buffer zone.
- This process was repeated for each lake.

Data Analysis

- Time series of total phosphorus and total nitrogen from 1976-2019 were created.¹⁰
- Pearson correlation coefficients between % area of both land cover types and concentration of TP and TN at each buffer distance across all lakes together, and their associated p-values were calculated using Excel.

RESULTS

1) Nutrient Levels

Total Phosphorus

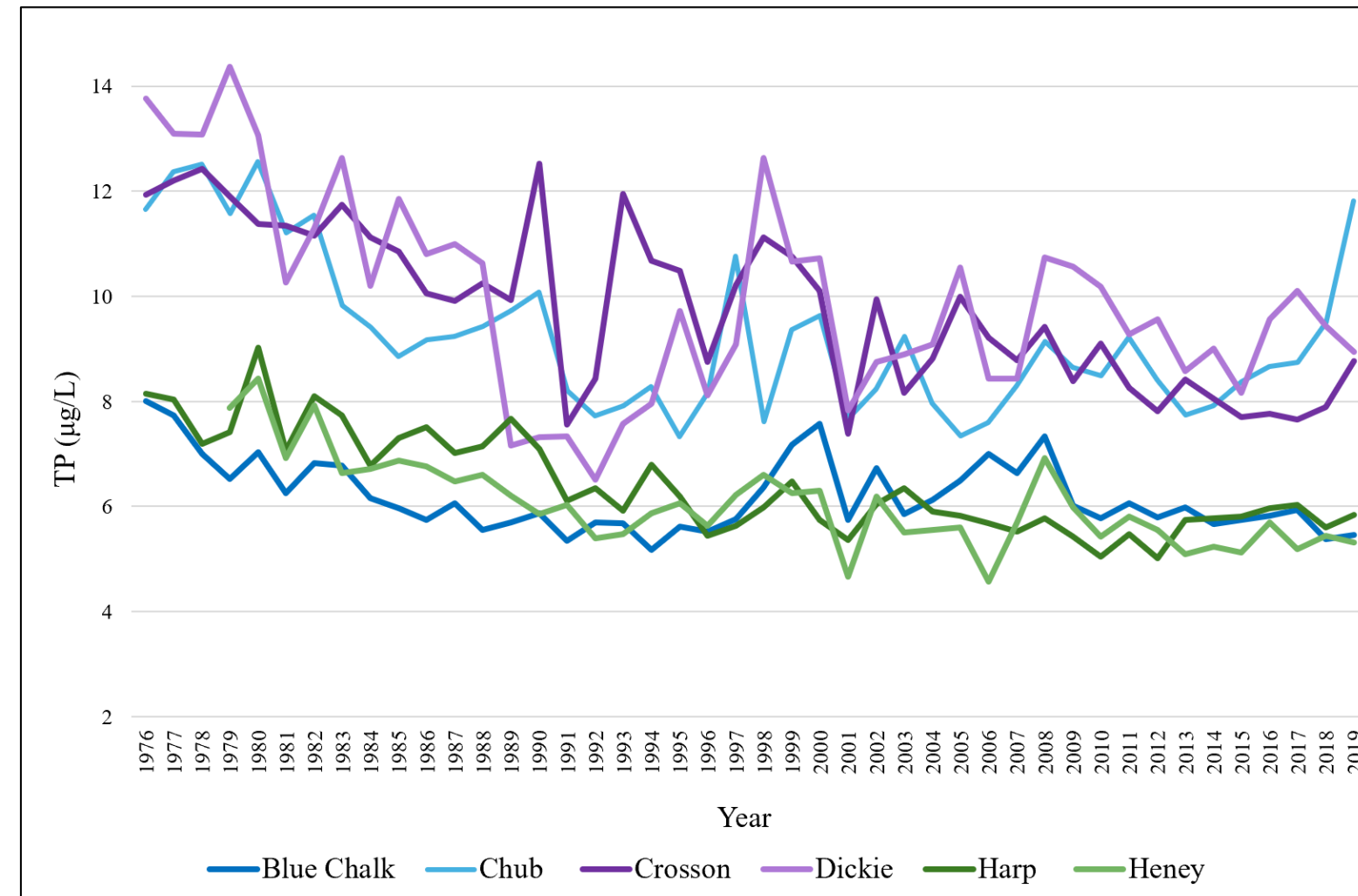


Figure 3. Ice-free mean annual TP concentrations (µg/L) of the six lakes from 1976-2019.¹⁰

Total Nitrogen

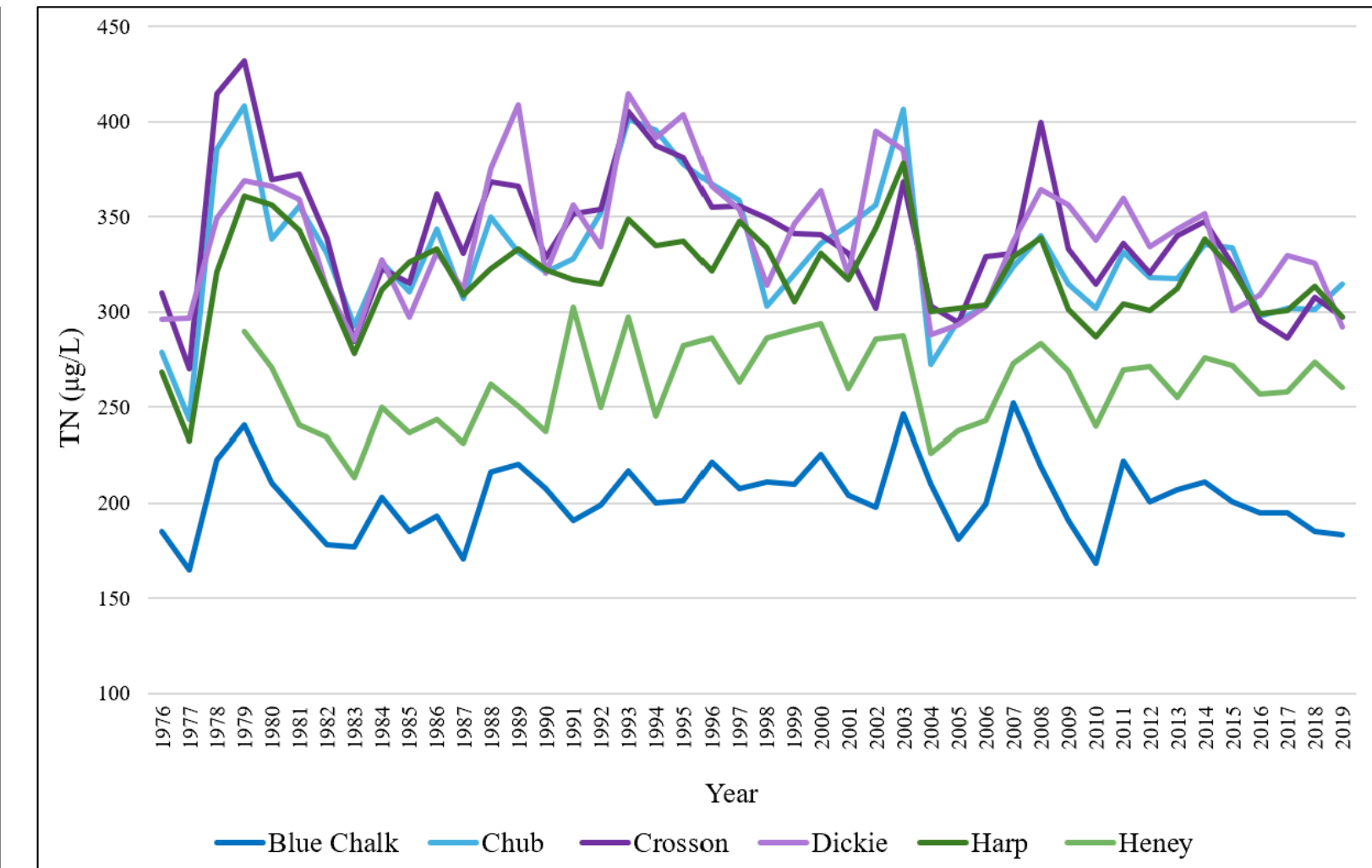


Figure 4. Ice-free mean annual TN concentrations (µg/L) of the six lakes from 1976-2019.¹⁰

2) Land Use

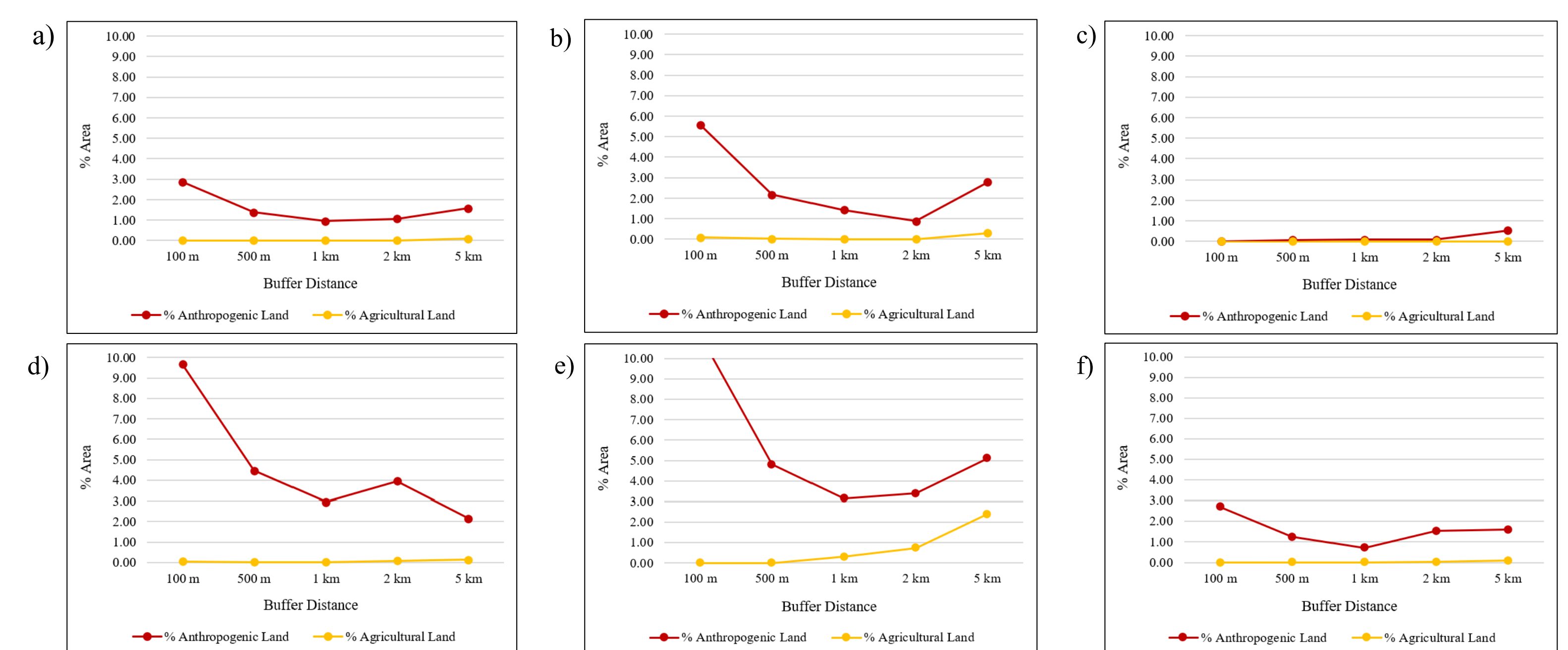


Figure 5. Percent area of anthropogenic (red) and agricultural (yellow) land use at 100m, 500m, 1km, 2km and 5km buffer zones around each lake. A) Represents Blue Chalk Lake, b) Chub Lake, c) Crosson Lake, d) Dickie Lake, e) Harp Lake and f) Heney Lake.

3) Correlation by Buffer Distance

Table 1. Pearson correlation coefficients (r) and p-values for relationships between land use (% area anthropogenic and agricultural) and TP and TN concentrations (µg/L) in 2019 at each of the 5 different buffer zones (100m-5km) across all 6 lakes together. P-value <0.05 would indicate a significant correlations between land use and nutrient concentrations.

Buffer Distance	Anthropogenic LU & TP Coefficient (r)	P-value	Anthropogenic LU & TN Coefficient (r)	P-value	Agricultural LU & TP Coefficient (r)	P-value	Agricultural LU & TN Coefficient (r)	P-value
100 m	0.0329	0.9582	0.3433	0.5717	0.8692	0.0557	0.4968	0.3946
500 m	-0.0115	0.9853	0.3024	0.6209	0.6411	0.2438	0.4330	0.4664
1 km	0.0024	0.9969	0.2986	0.6256	-0.3328	0.5842	0.2596	0.6733
2 km	-0.1805	0.7714	0.1897	0.7600	-0.3423	0.5728	0.2581	0.6751
5km	-0.1000	0.8729	0.3083	0.6138	-0.2830	0.6445	0.2671	0.6640

KEY FINDINGS

- TP concentrations declined in all lakes from 1976-2019, while TN increased in Blue Chalk and Heney Lakes.
- Chub, Dickie, and Crosson Lakes have the highest recent TP concentrations, while Chub, Crosson, and Harp Lakes have the highest recent TN concentrations.
- Anthropogenic land use is highest near lakes, with Harp and Dickie Lakes having the greatest percentages of surrounding area.
- Agricultural land use is generally low across all lakes but increases slightly with distance, with Harp and Chub Lakes having the highest percentages.
- No statistically significant relationships were found between land use and nutrient concentrations.
- TP and agricultural land use within a 100m buffer show the strongest correlation ($r = 0.8692$, $p = 0.0557$), indicating a potential positive relationship, though not statistically significant.
- Overall, results suggest that TP and TN concentrations in these lakes monitored by the DLMP appear mostly unaffected by urban and agricultural land use.

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